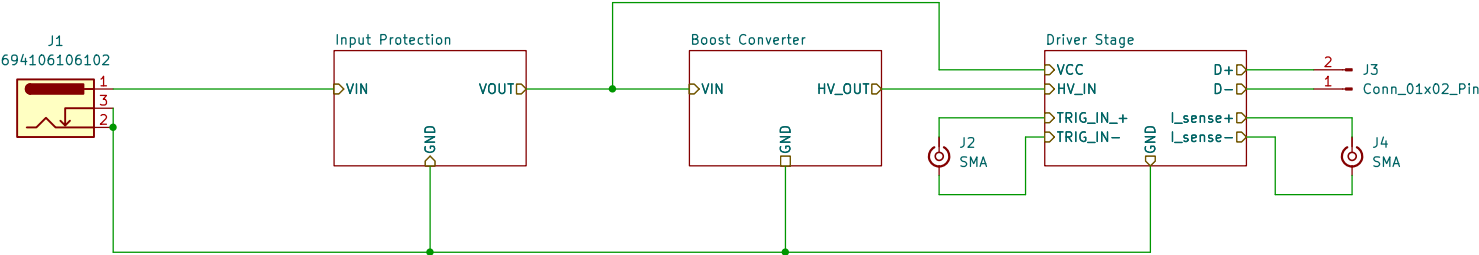
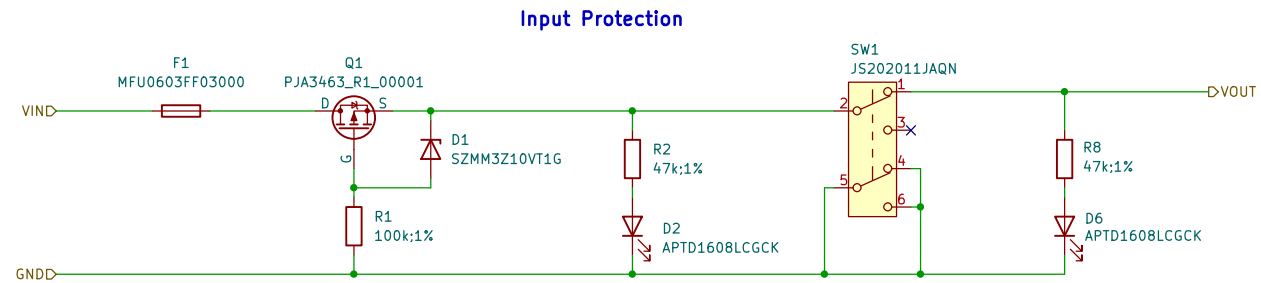


Avalanche Laser Driver



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Hannes Duske		
Sheet: /		
File: AvalancheLaserDriver.kicad_sch		
Title: Avalanche Laser Driver		
Size: A4	Date: 2026-02-27	Rev: 0.2
KiCad E.D.A. 10.0.4		Id: 1/4



Hannes Duske

Sheet: /Input Protection/

File: InputProtection.kicad_sch

Title: Avalanche Laser Driver

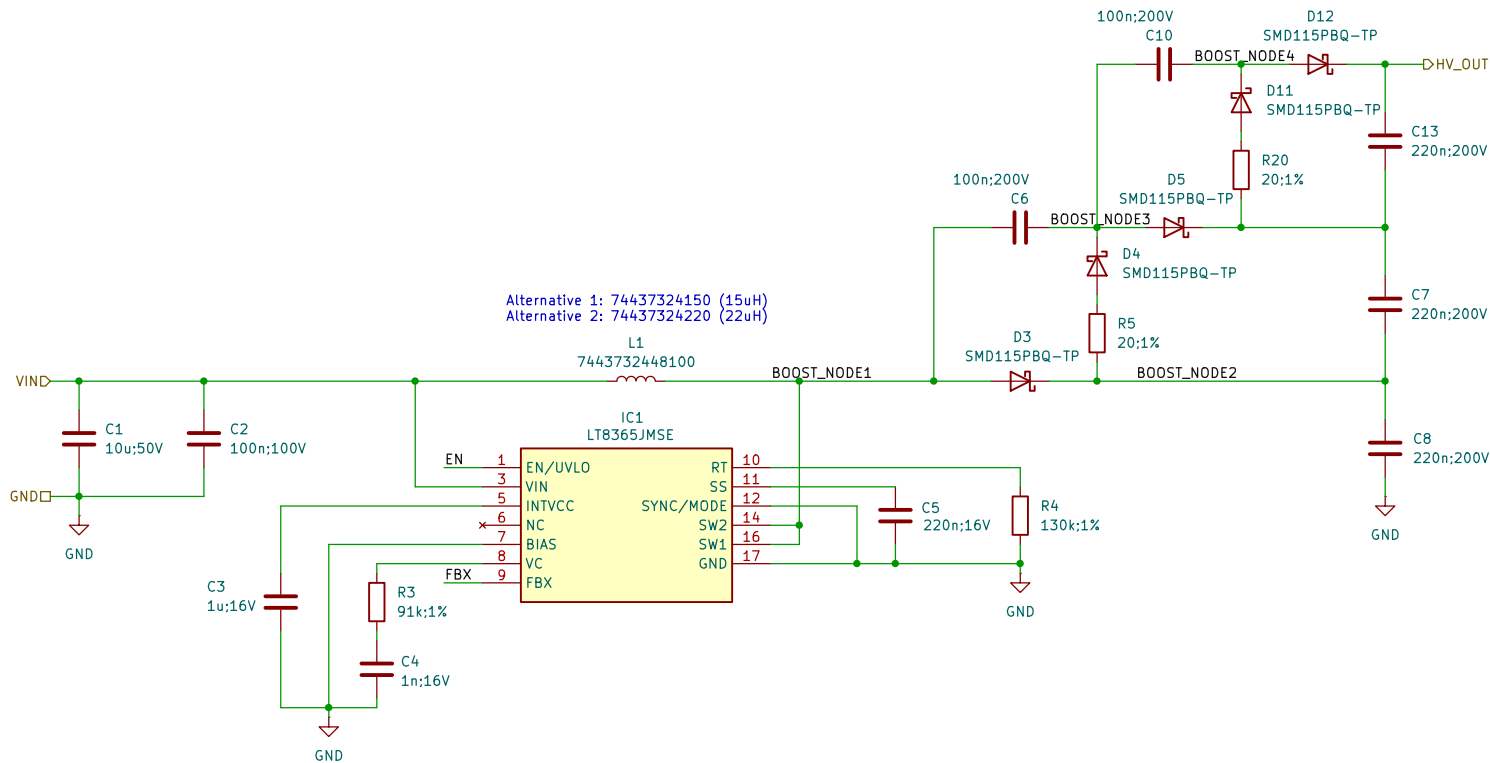
Size: A4

Date: 2026-02-27

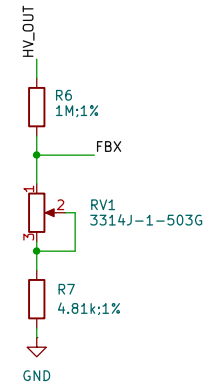
Rev: 0.2

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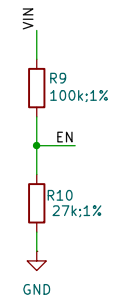
Id: 2/4



Feedback Network



Enable Network



The reference voltage of EN and FBX are 1.6V:
 - Cutoff input voltage is 7.5V with selected resistors on EN.
 - Output is theoretically adjustable from 165V to 334V with selected resistors on FBX.
 Measured output voltage with 5kΩ poti and 4.81kΩ fixed resistor is 160V to 316V.
 (Absolute max voltage rating of FM417 BJT is 320V.)

Poti RV1 terminal 2 and 3 tied together that the poti doesn't fail open circuit if wiper gets damaged.
 → limits VCC_HV in case of poti failure

RT=130kΩ compared to 107kΩ in other designs
 → f=330kHz, larger off-time-margin which would be near the limit at 400kHz,
 increased ripple doesn't matter because we're just charging the power caps with the output

Hannes Duske

Sheet: /Boost Converter/
 File: BoostConverter.kicad_sch

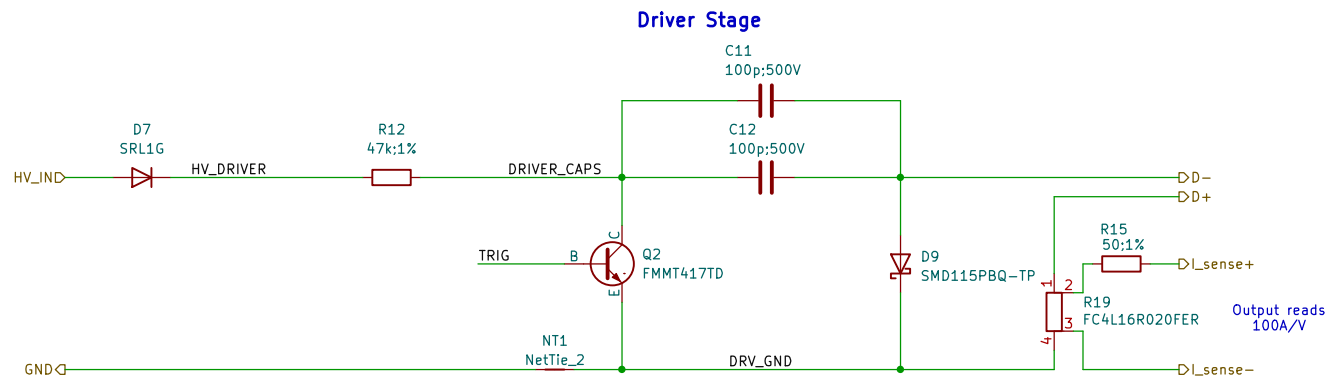
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Size: A4 Date: 2026-02-27

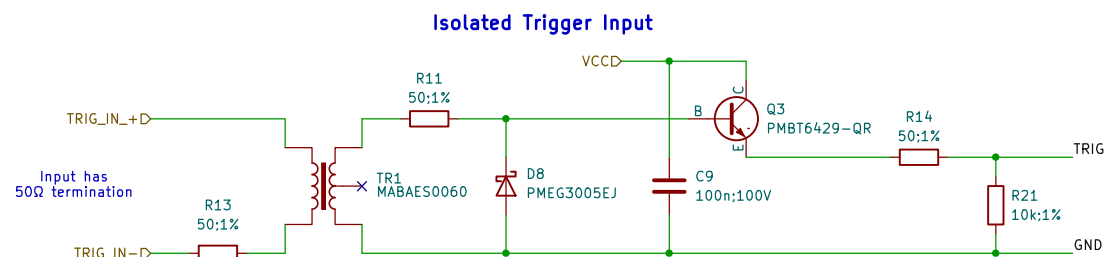
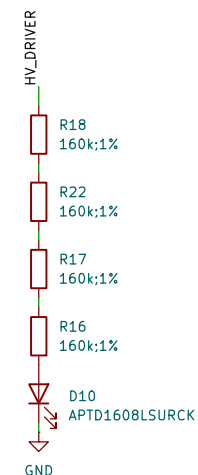
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Rev: 0.2

Id: 3/4



Status LED



The output pulse length roughly follows: $C \cdot V = I \cdot t$

- Reference: Les Lab produced a 2ns electrical pulse with 376pF at 200V
- At a constant pulse energy smaller caps with higher charging voltage yield shorter pulses than large caps with a lower voltage
- We will use 200pF up to 316V

Optimize design for small ESI to deliver the stored energy faster

- No pulse shaping resistors
- Smallest possible cap package with sufficient voltage rating
- Multiple caps in parallel with less capacitance for overall lower ESI
- Optimize the layout for smallest possible current loop
- (This design has all components on one side, double sided assembly could potentially have lower ESI because of smaller current loop between parallel traces)

Would have liked to use four 47pF caps, but couldn't find any that were in stock with 500V voltage rating.

A wideband RF transformer is used for galvanic isolation of the trigger input to protect the connected signal source. This also limits the trigger pulse width (high pass) which prevents damage to the system from sustained input signals that would keep the BJT turned on. The additional diode limits reverse voltage at the BJT base that happens when the trigger signal goes from high to low.

The BJT Q3 is added to achieve consist trigger signals at the avalanche transistor base regardless of the voltage of TRIG_IN. The addition of the trigger transistor considerably improved the stability of the output current pulses.

Hannes Duske

Sheet: /Driver Stage/
File: DriverStage.kicad_sch

Title: Avalanche Laser Driver

Size: A4 Date: 2026-02-27

KiCad E.D.A. 10.0.4

Rev: 0.2

Id: 4/4